YEAR 7 MATHS · CHAPTER 6 · MIND MAP

Negative numbers

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1 Introduction to negative integers

Numbers get smaller as you move left on the number line.

$-3 < 1 \quad -7 < -2$

4 degrees below zero is −4°C

In order: −8, −3, 0, 2

The opposite of 5 is −5

A lift at floor 2 goes down 5 floors: floor **−3**

2 Adding and subtracting positives

Adding moves right. Subtracting moves left.

$-3 + 5 = 2 \quad -2 + 6 = 4$

$2 - 7 = -5 \quad -4 - 3 = -7$

It's −3°C and warms by 8°: **5°C**

A diver at −12 m rises 5 m: **−7 m**

3 Adding and subtracting negatives

Adding a negative means subtract. Subtracting a negative means add.

$3 + (-5) = -2 \quad -1 + (-4) = -5$

$2 - (-3) = 5 \quad -6 - (-2) = -4$

Owing $5, then owing $4 more: $-5 + (-4) = -9$

Watch out: $+(-)$ acts like $-$, and $-(-)$ acts like $+$.

4 Multiplying and dividing

Same signs give a positive. Different signs give a negative.

$-3 \times 4 = -12 \quad -5 \times (-2) = 10$

$-20 \div 4 = -5 \quad -18 \div (-3) = 6$

Lose 3 points in each of 4 rounds: $-3 \times 4 = -12$

$-2 \times (-3) \times 4 = 24$

Watch out: this rule is for × and ÷ only: $-3 + (-2) = -5$.

5 Order of operations

Brackets, then × and ÷, then + and −, from left to right.

$-2 + 3 \times (-4) = -2 - 12 = -14$

$(-2 + 3) \times (-4) = -4$

$10 \div (-2) - 3 = -8$

$-3 \times 2 + 8 = -6 + 8 = 2$

Do the multiplying and dividing before the adding and subtracting.

6 Substituting integers

Put negative numbers in brackets when you substitute.

$a = -2,\ b = 3$: $a + b = 1 \quad ab = -6$

$b - a = 3 - (-2) = 5 \quad a^2 = 4$

$2a - b = -4 - 3 = -7$

$ab + 10 = -6 + 10 = 4$

Watch out: $(-2)^2 = 4$, not -4.

7 The Cartesian plane

Four quadrants, numbered anticlockwise from the top right. (x, y): across, then up or down.

$(-3, 2)$ is in the second quadrant

$(-1, -4)$ is in the third quadrant

$(2, -3)$ is in the fourth quadrant

Right 5 from $(-3, 2)$ gives $(2, 2)$

$(-2, 0)$ is on the x-axis, so it isn't in any quadrant.